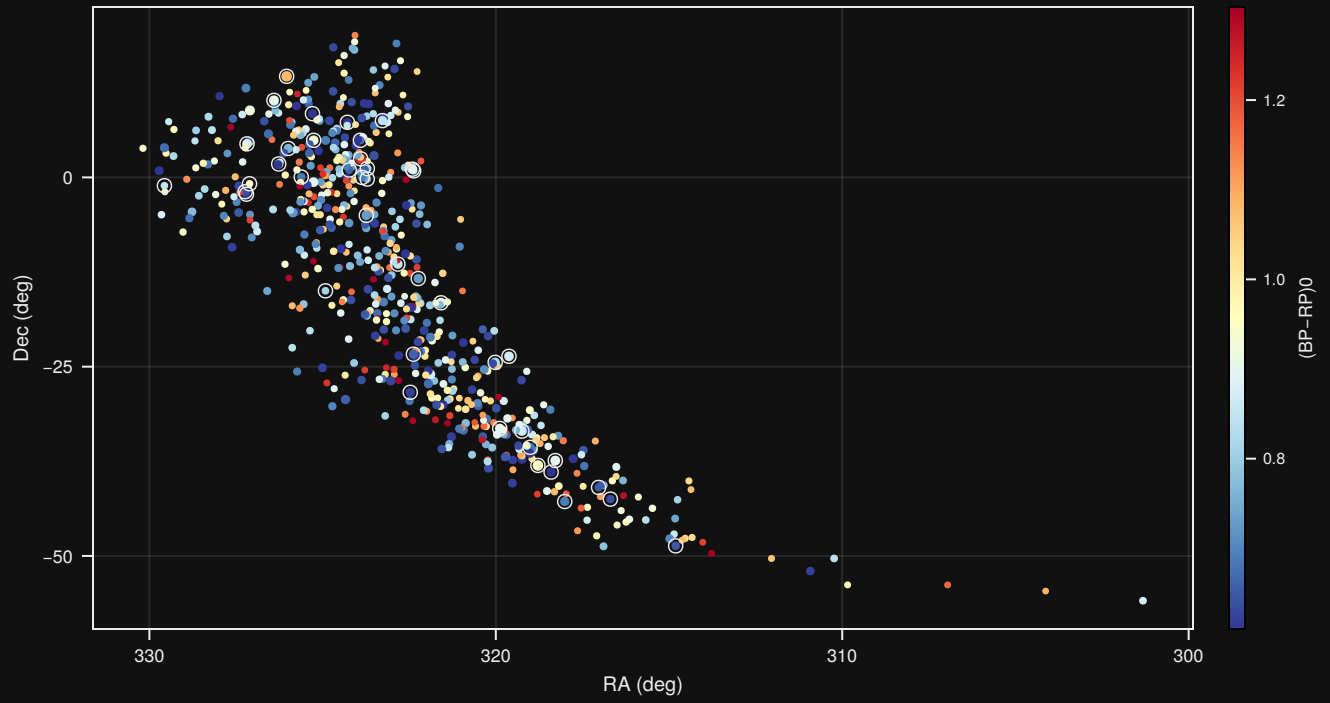
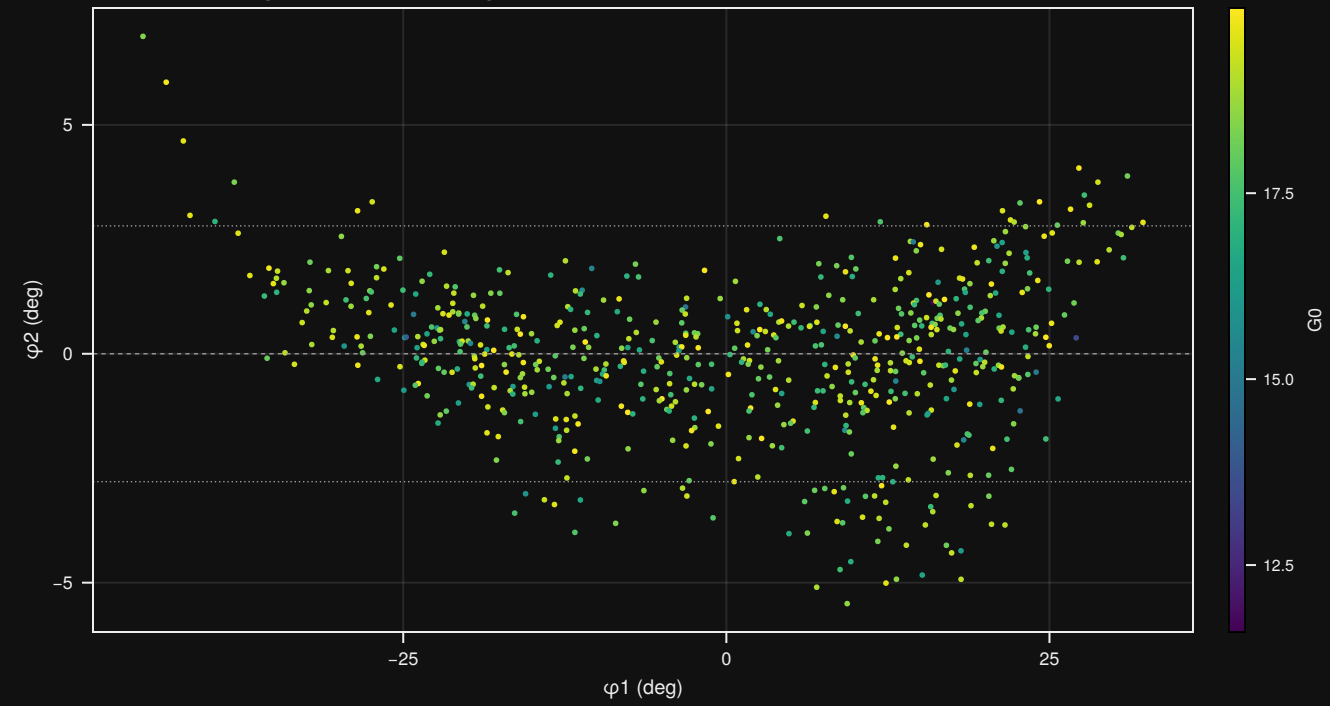
Phlegethon

MEMBERSHIP	
Stars in catalog	632
with Gaia GSP-Phot	363 (57%)
with radial velocity	41
Stream id (sID)	75
GEOMETRY · great-circle fit	
Centroid RA, Dec	322.8, -11.2 deg
Centroid l, b	37.7, -41.0 deg
Pole RA, Dec	234.7, 7.0 deg
Length (φ1, 2–98%)	62.1 deg
Width σ(φ2), robust	1.40 deg
KINEMATICS	
pmRA* med ± σ	-1.06 ± 2.59 mas/yr
pmDE med ± σ	-33.80 ± 5.11 mas/yr
μφ1* med ± σ	-33.91 ± 5.22 mas/yr
μφ2 med ± σ	-3.09 ± 1.31 mas/yr
Parallax (cat.) med	0.284 ± 0.198 mas
Parallax (wtd, ZP-corr.)	0.291 ± 0.011 mas
Distance	d = 3.4 ± 0.1 kpc
Physical length	3.73 kpc
RV median ± σ(MAD)	-276.8 ± 142.3 km/s
RV gradient dV/dφ1	-9.63 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	11.6–20.0 (18.6)
(BP−RP)0 median	0.87
GSP-Phot [M/H] (n=363)	-1.85 ± 0.48
GSP-Phot Teff median	5722 K
RUWE > 1.4	0.0 %
PM clusters (largest)	43 (25 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	632 | 632
n_RV catalog | paper	41 | 41
<[Fe/H]> paper	-2.19
Stellar mass (paper)	1700 Msun
Pole offset vs paper	1.4 deg

Sky map · catalog members (n = 632) · size ∝ brightness · ring = has RV

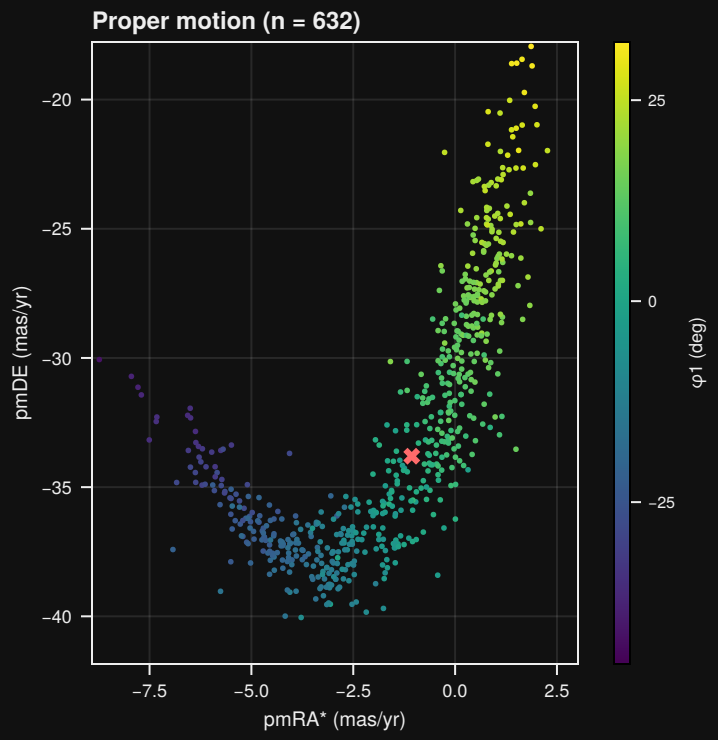


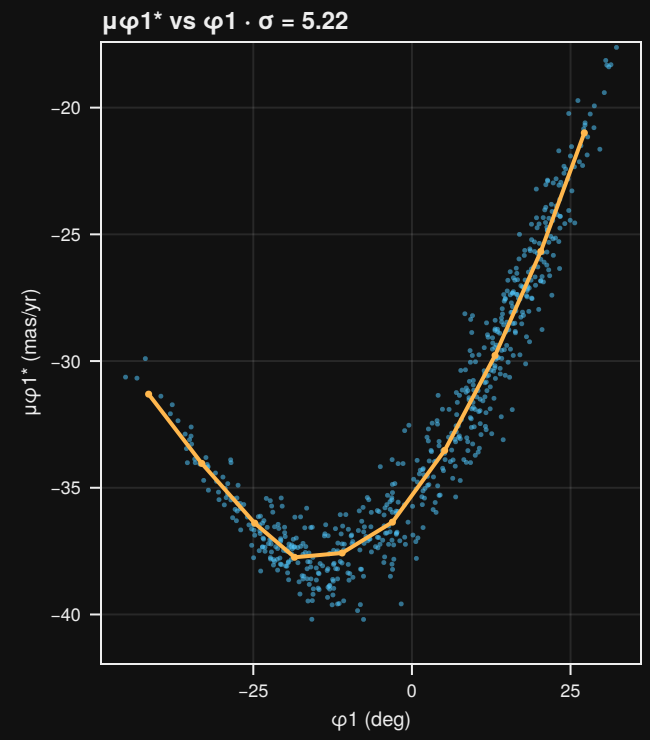
Stream frame (great-circle fit) · length 62.1° · width σ 1.40°

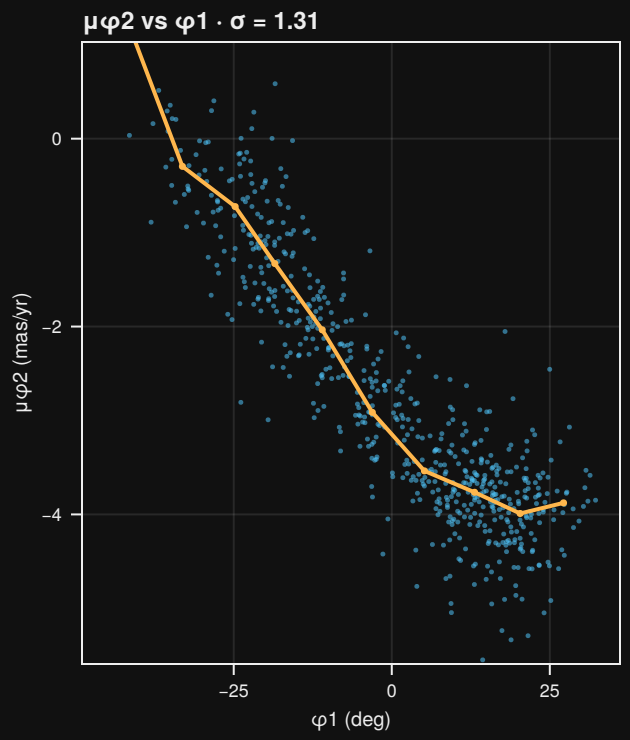


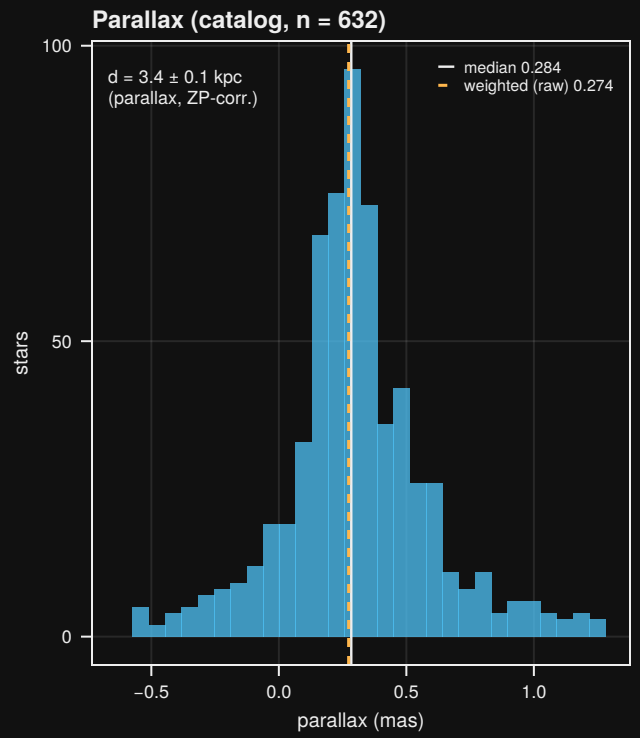
Phlegethon

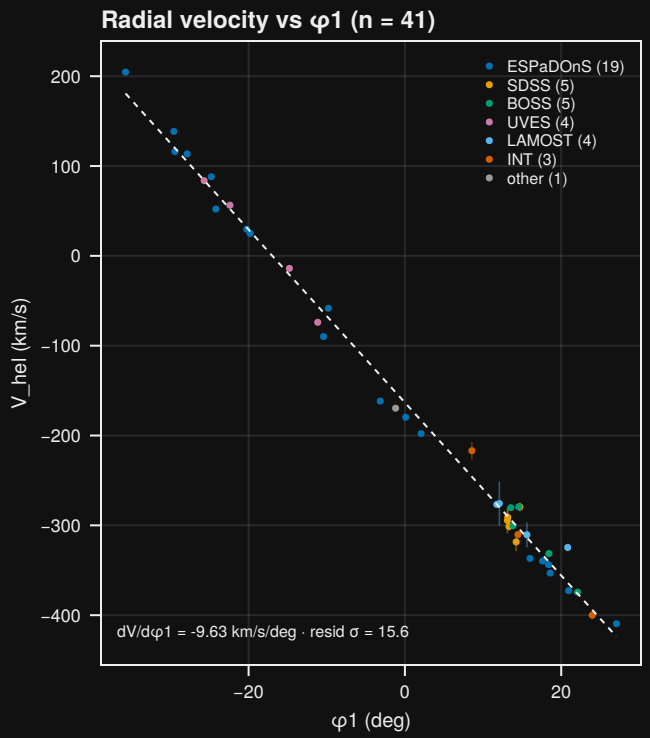
Kinematics · proper motion · parallax · radial velocity

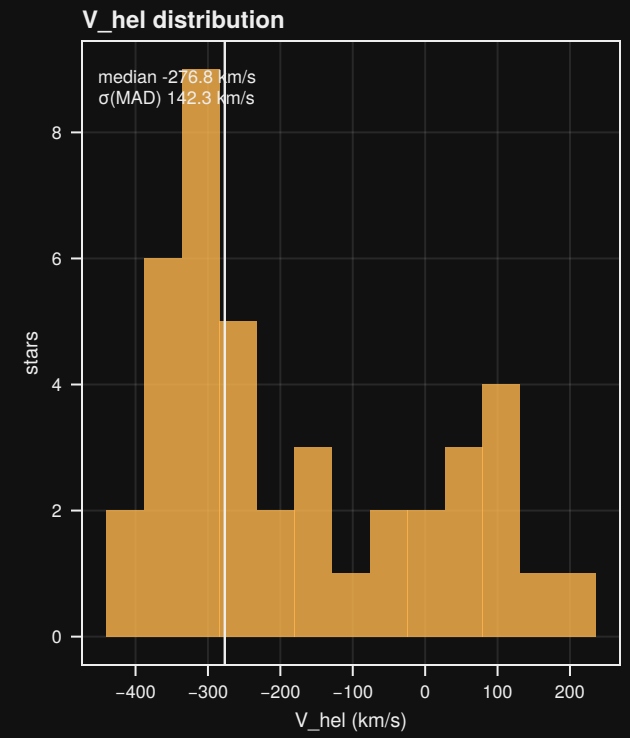






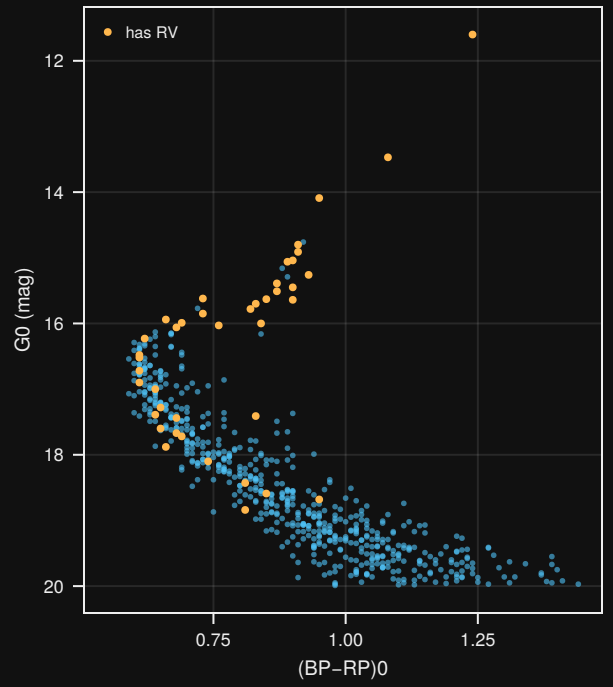




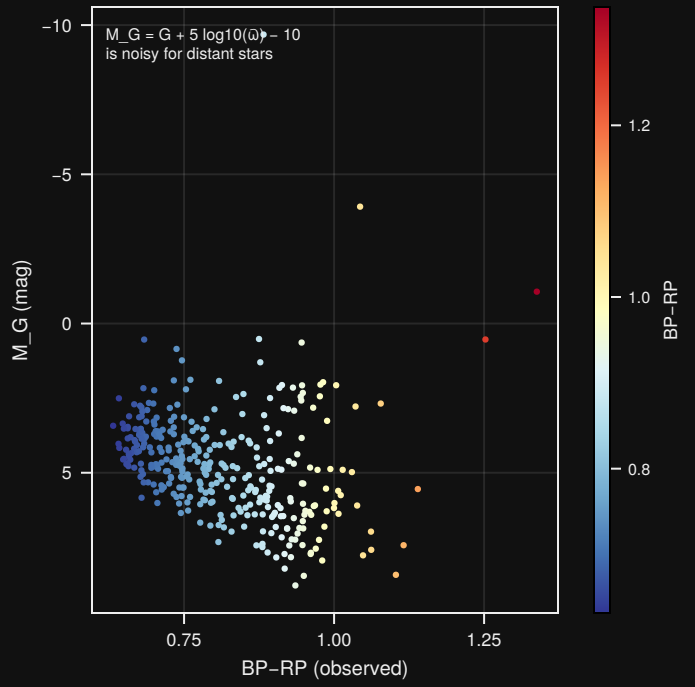


Phlegethon

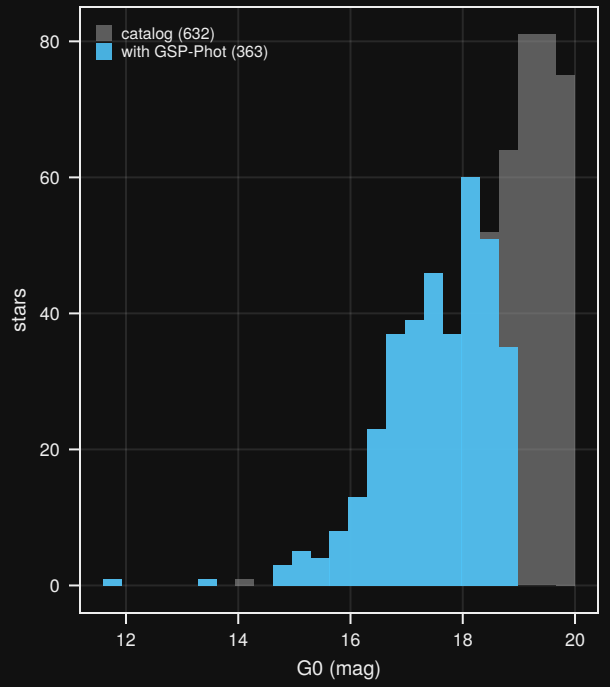
CMD · dereddened (n = 632)



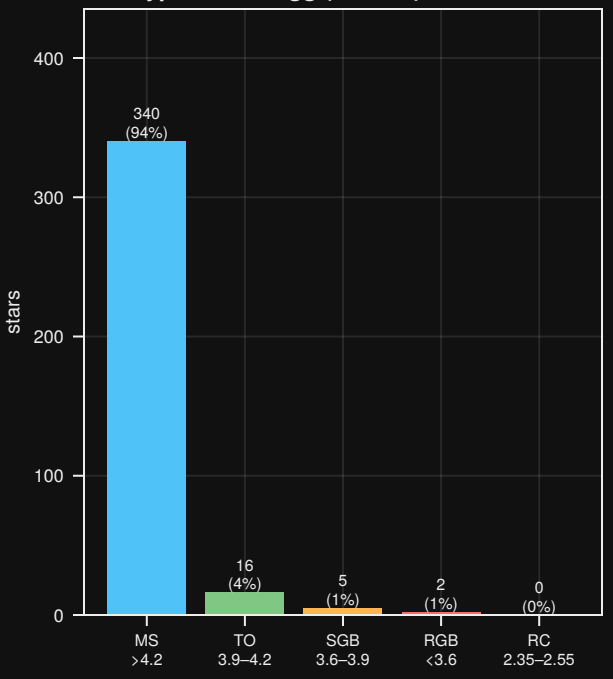
HR diagram · ϖ>0 (n = 350/363)



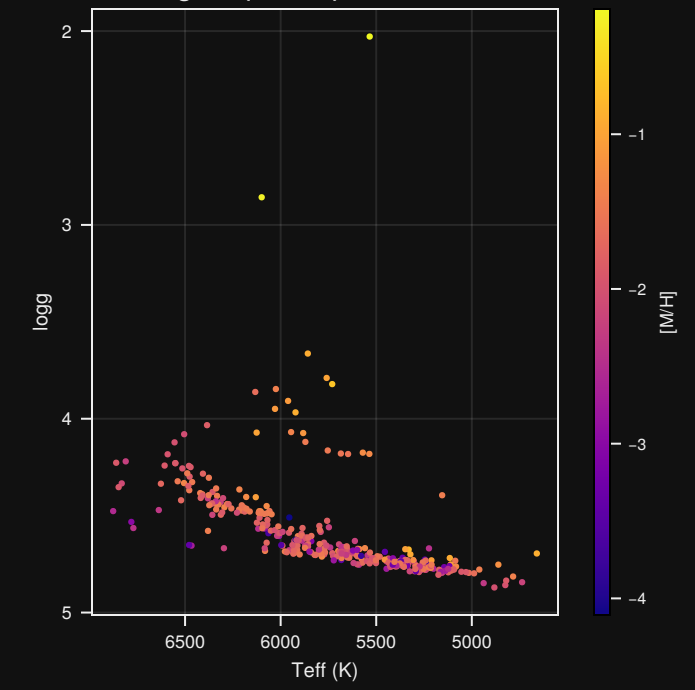
G0 & GSP-Phot coverage



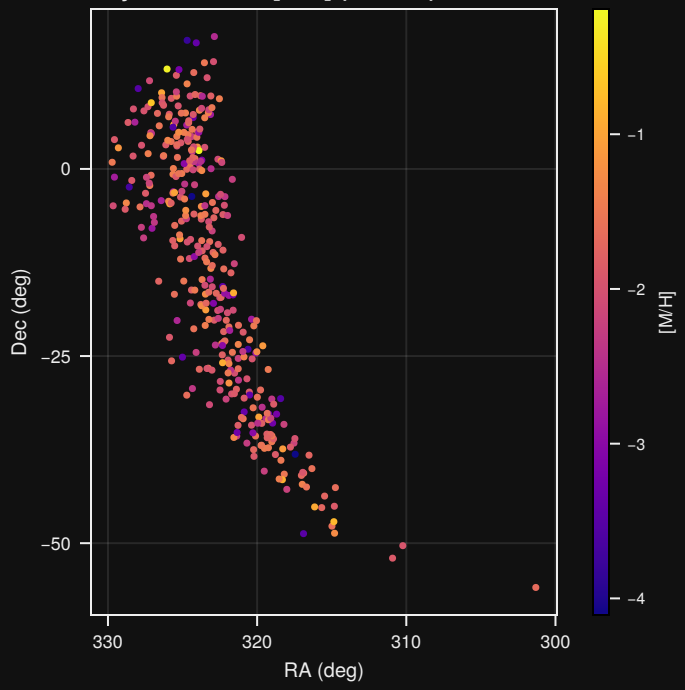
Star types from logg (n = 363)



Kiel diagram (n = 363)



Sky · GSP-Phot [M/H] (n = 363)

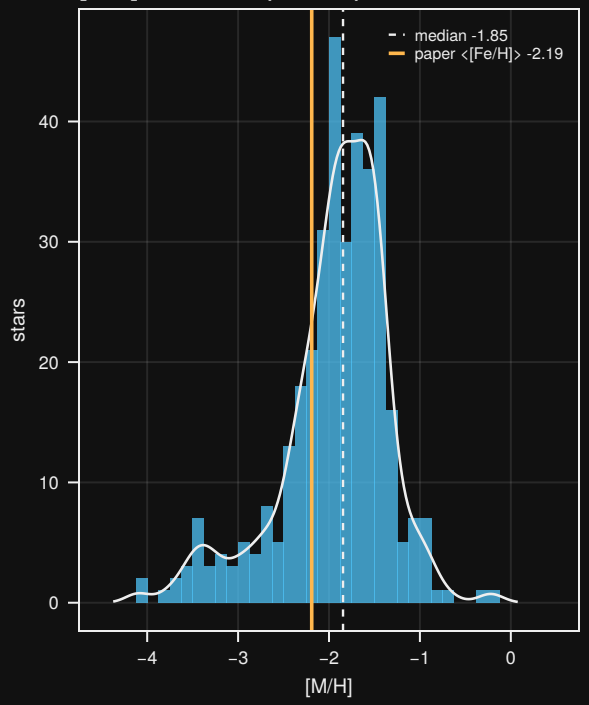


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≳ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

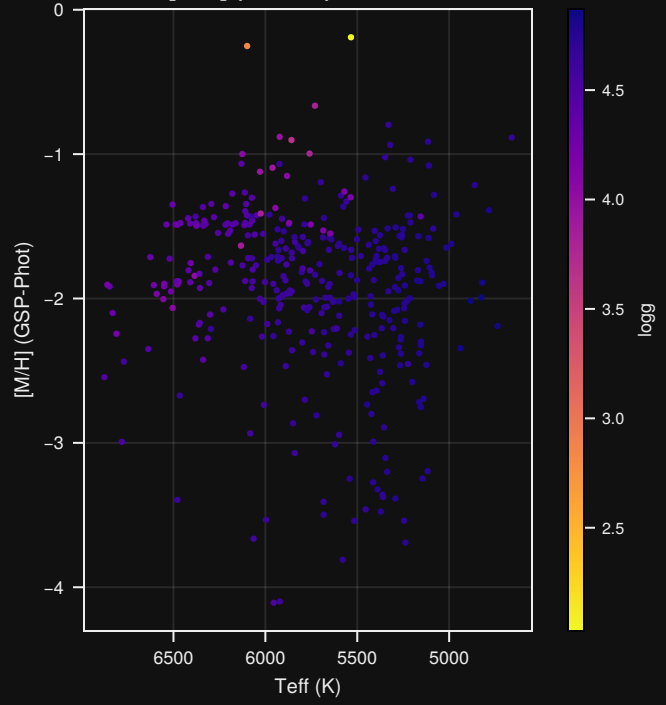
Phlegethon

Metallicity · age · astrometric quality

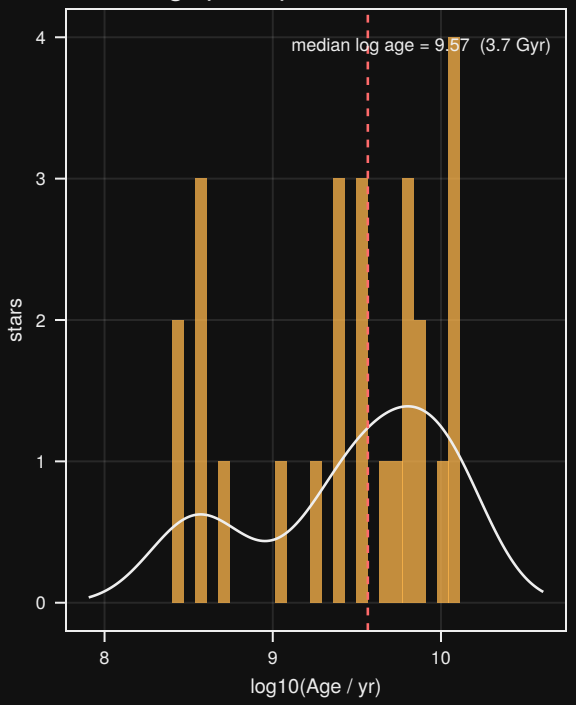
[M/H] GSP-Phot (n = 363)



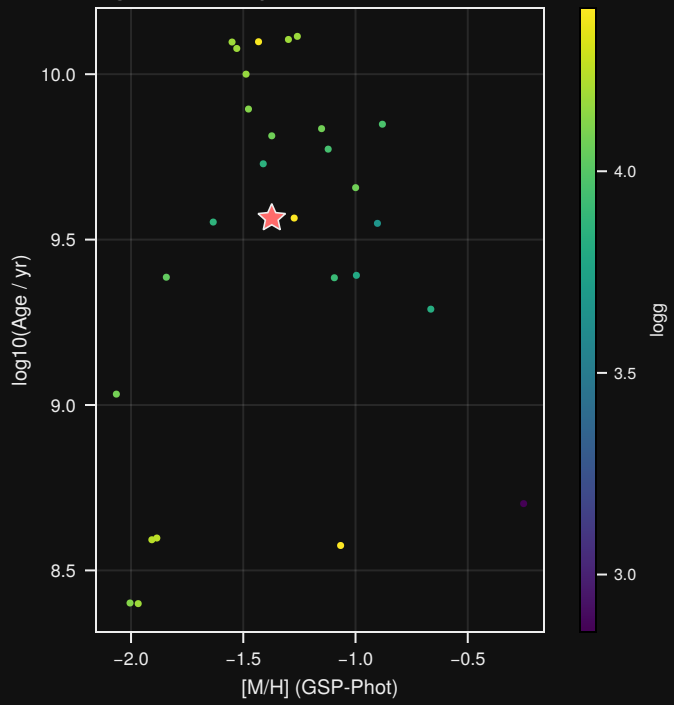
Teff vs [M/H] (n = 363)



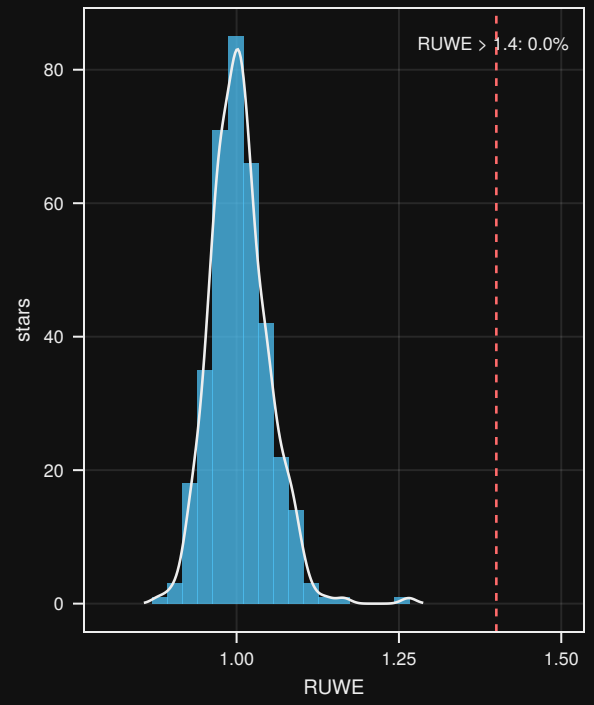
FLAME age (n = 27)



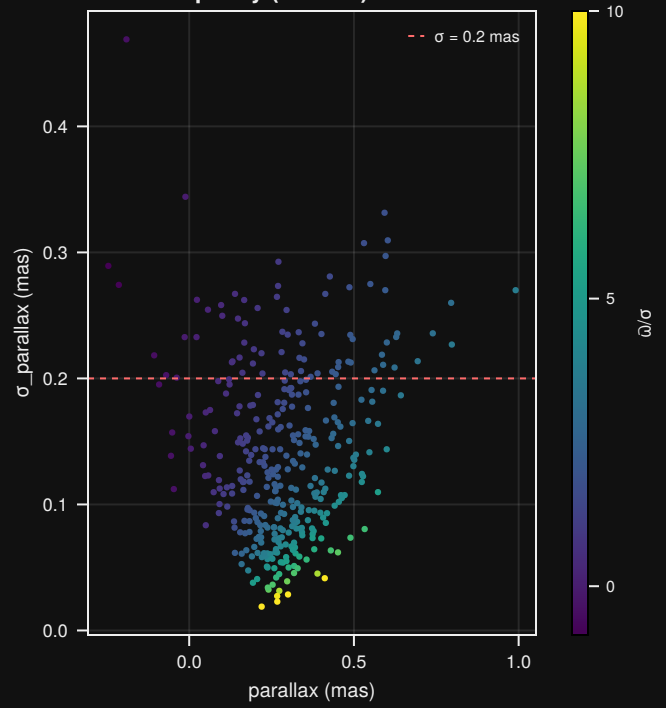
Age–metallicity (n = 27)



RUWE (n = 363)

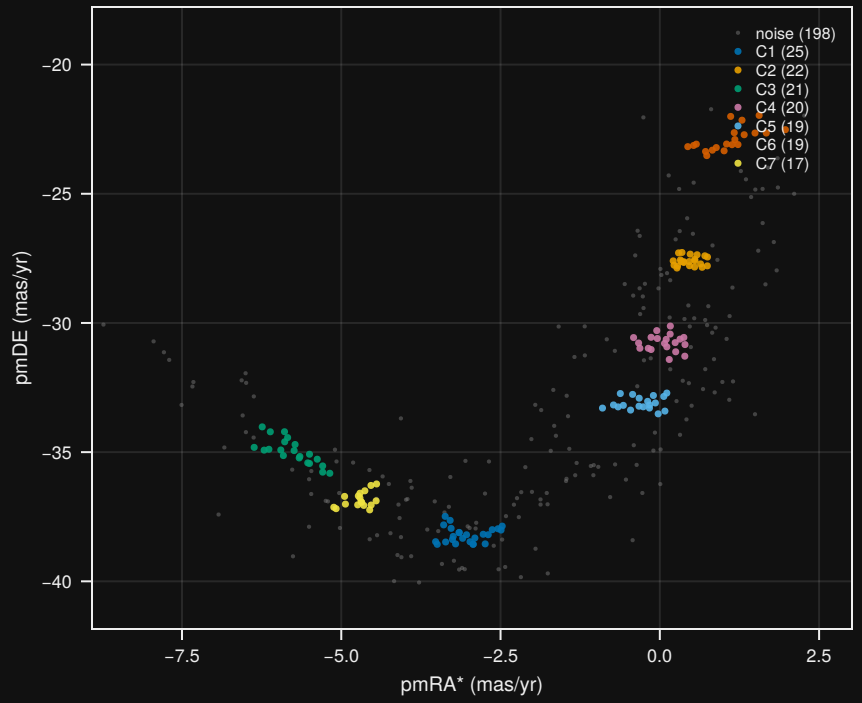


Parallax quality (n = 363)

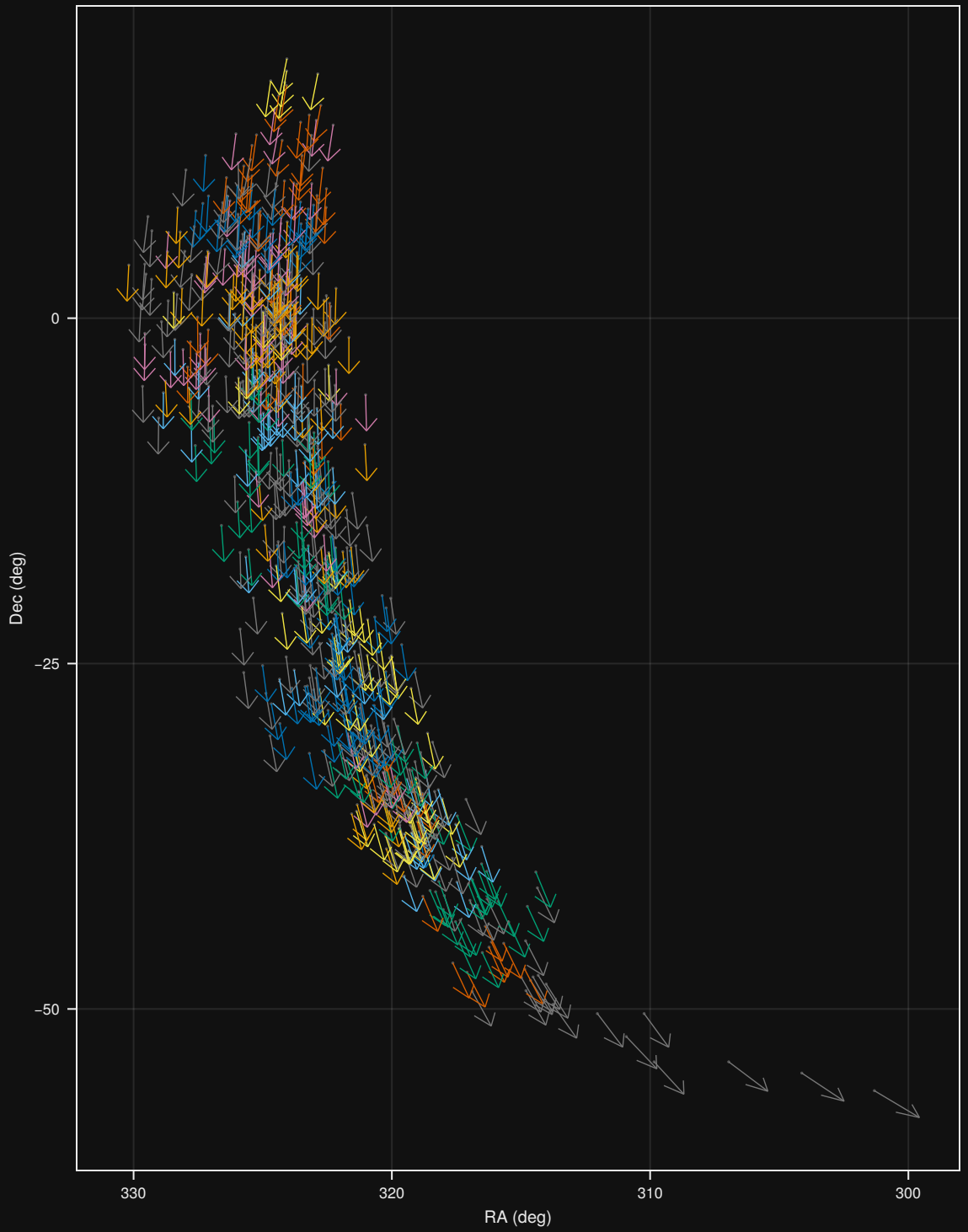


Phlegethon

PM clustering · HDBSCAN · 43 cluster(s) · noise 31%



Proper-motion directions (arrows) coloured by cluster



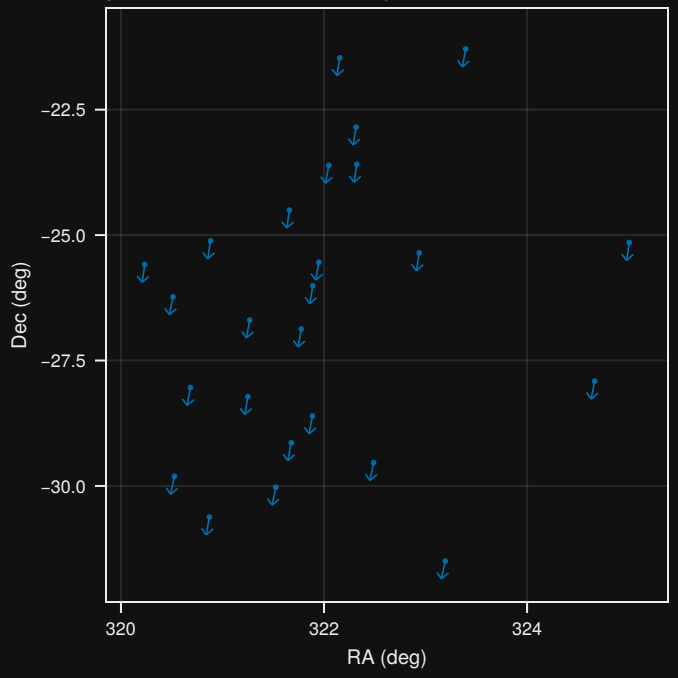
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	198	-1.46	-33.65	2.84	-1.88	11
C1	25	-3.10	-38.20	0.29	-1.91	0
C2	22	0.46	-27.62	0.20	-1.88	1
C3	21	-5.74	-34.94	0.36	-1.65	2
C4	20	0.11	-30.77	0.27	-2.28	4
C5	19	-0.26	-33.17	0.30	-1.83	0
C6	19	1.11	-23.07	0.43	-1.73	1
C7	17	-4.69	-36.92	0.20	-1.83	1
C8	17	0.90	-24.82	0.19	-1.75	2

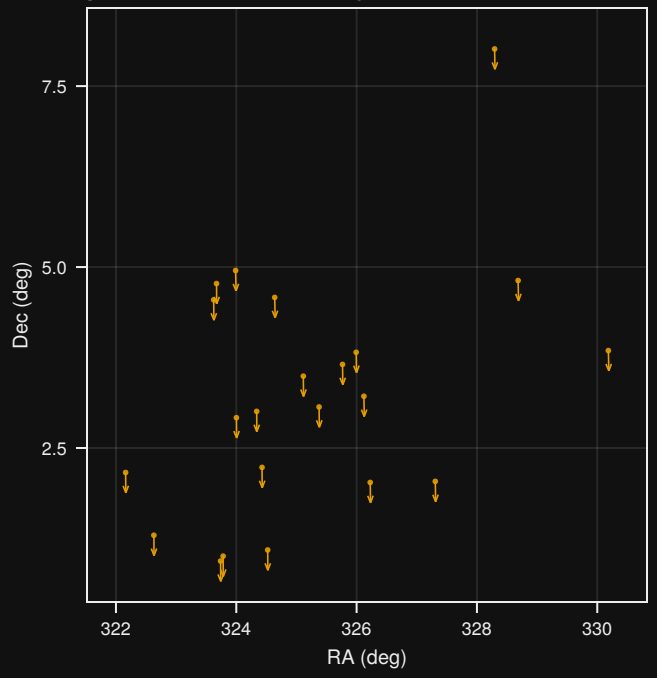
Phlegethon

Per-cluster proper-motion vector maps

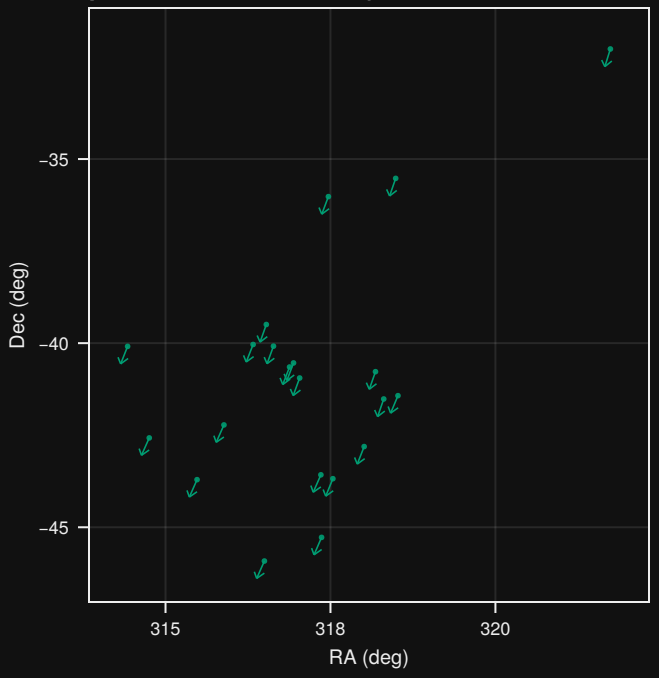
Cluster C1 · n = 25
pm = (-3.10, -38.20) mas/yr



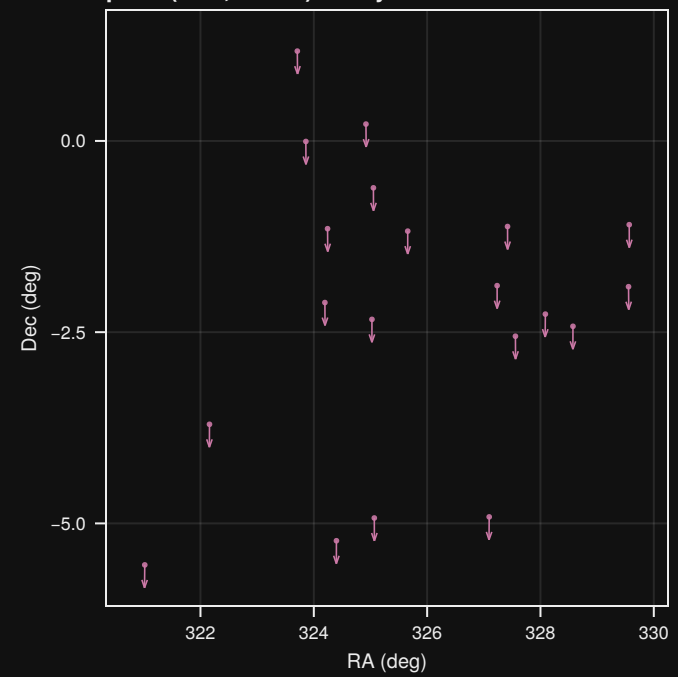
Cluster C2 · n = 22
pm = (0.46, -27.62) mas/yr



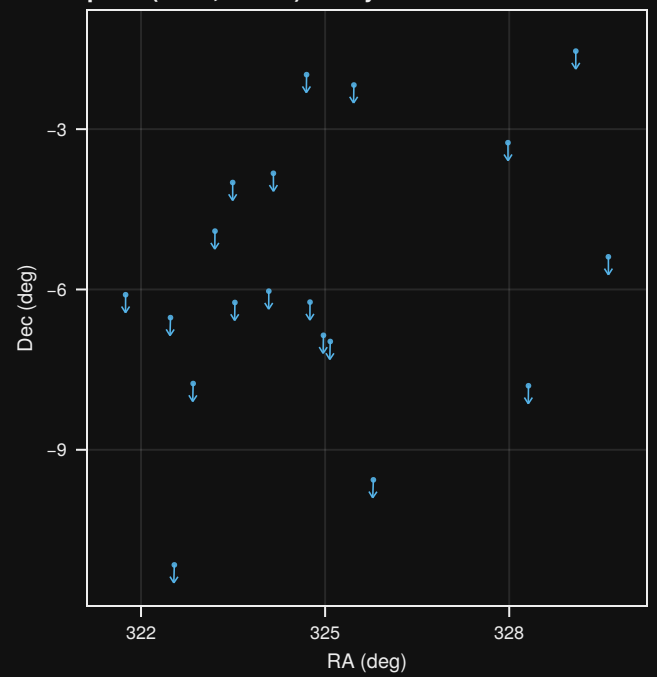
Cluster C3 · n = 21
pm = (-5.74, -34.94) mas/yr



Cluster C4 · n = 20
pm = (0.11, -30.77) mas/yr



Cluster C5 · n = 19
pm = (-0.26, -33.17) mas/yr



Noise · n = 198

